

$$X^{11}$$

$$\begin{aligned}
X^{11} = & \left\langle \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \right. \\
& \left(J_0(k\tau) J_0(l(\tau - \tau')) \right. \\
& + i \frac{v_\perp \cdot l_\perp}{l} J_0(k\tau) J_1(l(\tau - \tau')) \\
& + i \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_0(l(\tau - \tau')) \\
& \left. - \frac{v_\perp \cdot l_\perp}{l} \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_1(l(\tau - \tau')) \right) \\
& \left. \delta^{ij} \delta^{mn} [\alpha_1^i(u_\perp), \alpha_2^j(u_\perp)] [\alpha_1^m(v_\perp), \alpha_2^n(v_\perp)] \right\rangle
\end{aligned}$$

$$\begin{aligned}
X^{11} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left[J_0(k\tau) J_0(l(\tau - \tau')) \right. \\
& + i \frac{v_\perp \cdot l_\perp}{l} J_0(k\tau) J_1(l(\tau - \tau')) \\
& + i \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_0(l(\tau - \tau')) \\
& \left. - \frac{v_\perp \cdot l_\perp}{l} \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_1(l(\tau - \tau')) \right] \\
& \delta^{ij} \delta^{mn} f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

$$X^{1\beta}, \beta = 2, 3$$

$$\begin{aligned}
X^{1\beta} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left[-i\varepsilon^{\beta\beta'} \frac{l^{\beta'}}{l} J_0(k\tau) J_1(l(\tau - \tau')) \varepsilon^{mn} \right. \\
& - i \left(i\varepsilon^{\beta\beta'} v^{\beta'} J_0(l(\tau - \tau')) \varepsilon^{mn} + v^1 \frac{l^\beta}{l} J_1(l(\tau - \tau')) \delta^{mn} \right) J_0(k\tau) \\
& + \frac{v_\perp \cdot k_\perp}{k} \varepsilon^{\beta\beta'} \frac{l^{\beta'}}{l} J_1(k\tau) J_1(l(\tau - \tau')) \varepsilon^{mn} \\
& \left. + \frac{v_\perp \cdot k_\perp}{k} \left(i\varepsilon^{\beta\beta'} v^{\beta'} J_0(l(\tau - \tau')) \varepsilon^{mn} + v^1 \frac{l^\beta}{l} J_1(l(\tau - \tau')) \delta^{mn} \right) J_1(k\tau) \right] \\
& \delta^{ij} f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

$$\begin{aligned}
X^{1\beta} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left[\mp i \frac{l^{\beta'}}{l} J_0(k\tau) J_1(l(\tau - \tau')) \varepsilon^{mn} \right. \\
& - i \left(\pm i v^{\beta'} J_0(l(\tau - \tau')) \varepsilon^{mn} + v^1 \frac{l^\beta}{l} J_1(l(\tau - \tau')) \delta^{mn} \right) J_0(k\tau) \\
& \pm \frac{v_\perp \cdot k_\perp}{k} \frac{l^{\beta'}}{l} J_1(k\tau) J_1(l(\tau - \tau')) \varepsilon^{mn} \\
& \left. + \frac{v_\perp \cdot k_\perp}{k} \left(\pm i v^{\beta'} J_0(l(\tau - \tau')) \varepsilon^{mn} + v^1 \frac{l^\beta}{l} J_1(l(\tau - \tau')) \delta^{mn} \right) J_1(k\tau) \right] \\
& \delta^{ij} f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

(removing the terms with Levi-Civita in the nm space)

$$\begin{aligned}
X^{1\beta} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left[-i v^1 \frac{l^\beta}{l} J_0(k\tau) J_1(l(\tau - \tau')) \right. \\
& \left. + \frac{v_\perp \cdot k_\perp}{k} v^1 \frac{l^\beta}{l} J_1(k\tau) J_1(l(\tau - \tau')) \right] \\
& \delta^{ij} \delta^{mn} f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

$$X^{\alpha 1}, \alpha = 2, 3$$

$$\begin{aligned}
X^{\alpha 1} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left[-i\varepsilon^{\alpha\alpha'} \frac{k^{\alpha'}}{k} J_0(l(\tau - \tau')) J_1(k\tau) \varepsilon^{ij} \right. \\
& - i \left(i\varepsilon^{\alpha\alpha'} v^{\alpha'} J_0(k\tau) \varepsilon^{ij} + v^1 \frac{k^\alpha}{k} J_1(k\tau) \delta^{ij} \right) J_0(l(\tau - \tau')) \\
& + \frac{v_\perp \cdot l_\perp}{l} \varepsilon^{\alpha\alpha'} \frac{k^{\alpha'}}{k} J_1(l(\tau - \tau')) J_1(k\tau) \varepsilon^{ij} \\
& \left. + \frac{v_\perp \cdot l_\perp}{l} \left(i\varepsilon^{\alpha\alpha'} v^{\alpha'} J_0(k\tau) \varepsilon^{ij} + v^1 \frac{k^\alpha}{k} J_1(k\tau) \delta^{ij} \right) J_1(l(\tau - \tau')) \right] \\
& \delta^{mn} f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

(removing the terms with Levi-Civita in the ij space)

$$\begin{aligned}
X^{\alpha 1} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left[-iv^1 \frac{k^\alpha}{k} J_1(k\tau) J_0(l(\tau - \tau')) \right. \\
& \left. + \frac{v_\perp \cdot l_\perp}{l} v^1 \frac{k^\alpha}{k} J_1(k\tau) J_1(l(\tau - \tau')) \right] \\
& \delta^{ij} \delta^{mn} f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

$$X^{\alpha\beta}, \alpha, \beta = 2, 3$$

$$\begin{aligned}
X^{\alpha\beta} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left\{ \left[\left(\delta^{\alpha\beta} \frac{l_\perp \cdot k_\perp}{l k} - \frac{l^\alpha k^\beta}{l k} \right) J_1(k\tau) J_1(l(\tau - \tau')) \right. \right. \\
& - i \left(\delta^{\alpha\beta} \frac{v_\perp \cdot k_\perp}{k} - \frac{v^\alpha k^\beta}{k} \right) J_1(k\tau) J_0(l(\tau - \tau')) \\
& - i \left(\delta^{\alpha\beta} \frac{l_\perp \cdot v_\perp}{l} - \frac{l^\alpha v^\beta}{l} \right) J_0(k\tau) J_1(l(\tau - \tau')) \\
& \left. + \left(\delta^{\alpha\beta} v_\perp^2 - v^\alpha v^\beta \right) J_0(k\tau) J_0(l(\tau - \tau')) \right] \varepsilon^{ij} \varepsilon^{mn} \\
& - (v^1)^2 \frac{k^\alpha l^\beta}{k l} J_1(k\tau) J_1(l(\tau - \tau')) \delta^{ij} \delta^{mn} \Big\} \\
& f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

$$\begin{aligned}
X^{\alpha\beta} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \left\{ \left[\left(\delta^{\alpha\beta} - \frac{l^\alpha k^\beta}{l_\perp \cdot k_\perp} \right) \frac{l_\perp \cdot k_\perp}{l k} J_1(k\tau) J_1(l(\tau - \tau')) \right. \right. \\
& - i \left(\delta^{\alpha\beta} - \frac{v^\alpha k^\beta}{v_\perp \cdot k_\perp} \right) \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_0(l(\tau - \tau')) \\
& - i \left(\delta^{\alpha\beta} - \frac{l^\alpha v^\beta}{l_\perp \cdot v_\perp} \right) \frac{l_\perp \cdot v_\perp}{l} J_0(k\tau) J_1(l(\tau - \tau')) \\
& \left. + \left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v_\perp^2} \right) v_\perp^2 J_0(k\tau) J_0(l(\tau - \tau')) \right] \varepsilon^{ij} \varepsilon^{mn} \\
& - (v^1)^2 \frac{k^\alpha l^\beta}{k l} J_1(k\tau) J_1(l(\tau - \tau')) \delta^{ij} \delta^{mn} \Big\} \\
& f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

ENVELOPE

$$\begin{aligned}
X^{\alpha\beta} = & \frac{g^4}{2N_c} \int_0^\tau d\tau' \int \frac{d^2 k_\perp}{(2\pi^2)} \int \frac{d^2 l_\perp}{(2\pi^2)} \int d^2 u_\perp \int d^2 v_\perp e^{i(x_\perp - u_\perp)k_\perp} e^{i(y_\perp - v_\perp)l_\perp} \\
& \dots \\
& f^{bca} f^{dea} \langle \alpha_{1b}^i(u_\perp) \alpha_{1d}^m(v_\perp) \rangle \langle \alpha_{2c}^j(u_\perp) \alpha_{2e}^n(v_\perp) \rangle
\end{aligned}$$

FULL TENSOR (without integral/colour space envelope)

$$\begin{aligned}
X^{11} = & \dots \left[J_0(k\tau) J_0(l(\tau - \tau')) \right. \\
& + i \frac{v_\perp \cdot l_\perp}{l} J_0(k\tau) J_1(l(\tau - \tau')) \\
& + i \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_0(l(\tau - \tau')) \\
& \left. - \frac{v_\perp \cdot l_\perp}{l} \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_1(l(\tau - \tau')) \right] \delta^{ij} \delta^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
X^{1\beta} = & \dots \left[-i v^1 \frac{l^\beta}{l} J_0(k\tau) J_1(l(\tau - \tau')) \right. \\
& \left. + \frac{v_\perp \cdot k_\perp}{k} v^1 \frac{l^\beta}{l} J_1(k\tau) J_1(l(\tau - \tau')) \right] \delta^{ij} \delta^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
X^{\alpha 1} = & \dots \left[-i v^1 \frac{k^\alpha}{k} J_1(k\tau) J_0(l(\tau - \tau')) \right. \\
& \left. + \frac{v_\perp \cdot l_\perp}{l} v^1 \frac{k^\alpha}{k} J_1(k\tau) J_1(l(\tau - \tau')) \right] \delta^{ij} \delta^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
X^{\alpha\beta} = & \dots \left\{ \left[- \left(\delta^{\alpha\beta} - \frac{l^\alpha k^\beta}{l_\perp \cdot k_\perp} \right) \frac{l_\perp \cdot k_\perp}{l k} J_1(k\tau) J_1(l(\tau - \tau')) \right. \right. \\
& - i \left(\delta^{\alpha\beta} - \frac{v^\alpha k^\beta}{v_\perp \cdot k_\perp} \right) \frac{v_\perp \cdot k_\perp}{k} J_1(k\tau) J_0(l(\tau - \tau')) \\
& - i \left(\delta^{\alpha\beta} - \frac{l^\alpha v^\beta}{l_\perp \cdot v_\perp} \right) \frac{l_\perp \cdot v_\perp}{l} J_0(k\tau) J_1(l(\tau - \tau')) \\
& + \left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v_\perp^2} \right) v_\perp^2 J_0(k\tau) J_0(l(\tau - \tau')) \left. \right] \varepsilon^{ij} \varepsilon^{mn} \\
& \left. - (v^1)^2 \frac{k^\alpha l^\beta}{k l} J_1(k\tau) J_1(l(\tau - \tau')) \delta^{ij} \delta^{mn} \right\} \dots
\end{aligned}$$

$$\begin{aligned}
X^{\alpha\beta} = & \dots \left\{ \left[-\varepsilon^{\alpha\sigma} \frac{k^\sigma}{k} \varepsilon^{\beta\rho} \frac{l^\rho}{l} J_1(k\tau) J_1(l(\tau - \tau')) \right. \right. \\
& - i\varepsilon^{\alpha\sigma} \frac{k^\sigma}{k} \varepsilon^{\beta\rho} v^\rho J_1(k\tau) J_0(l(\tau - \tau')) \\
& - i\varepsilon^{\alpha\rho} v^\rho \varepsilon^{\beta\sigma} \frac{l^\sigma}{l} J_0(k\tau) J_1(l(\tau - \tau')) \\
& \left. \left. + \varepsilon^{\alpha\sigma} v^\sigma \varepsilon^{\beta\rho} v^\rho J_0(k\tau) J_0(l(\tau - \tau')) \right] \varepsilon^{ij} \varepsilon^{mn} \right. \\
& \left. - (v^1)^2 \frac{k^\alpha l^\beta}{k l} J_1(k\tau) J_1(l(\tau - \tau')) \delta^{ij} \delta^{mn} \right\} \dots
\end{aligned}$$

CONTRACTIONS (without integral/colour space envelope)

$$\begin{aligned}
v^\alpha v^\beta X^{\alpha\beta} = & \dots (v^\perp)^2 J_0(k\tau) J_0(l(\tau - \tau')) \delta^{ij} \delta^{mn} \\
& - \left(v_\perp^2 \frac{(k_\perp \cdot l_\perp)}{k l} - \frac{(v_\perp \cdot k_\perp)}{k} \frac{(v_\perp \cdot l_\perp)}{l} \right) J_1(k\tau) J_1(l(\tau - \tau')) \varepsilon^{ij} \varepsilon^{mn} \dots \\
= & \dots (v^\perp)^2 J_0(k\tau) J_0(l(\tau - \tau')) \delta^{ij} \delta^{mn} \\
& - \left(v_\perp \times \frac{k_\perp}{k} \right) \left(v_\perp \times \frac{l_\perp}{l} \right) J_1(k\tau) J_1(l(\tau - \tau')) \varepsilon^{ij} \varepsilon^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} = & \dots (\delta^{ij} \delta^{mn} + v_\perp^2 \varepsilon^{ij} \varepsilon^{mn}) J_0(k\tau) J_0(l(\tau - \tau')) \\
& + i \frac{v_\perp \cdot l_\perp}{l} (\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0(k\tau) J_1(l(\tau - \tau')) \\
& + i \frac{v_\perp \cdot k_\perp}{k} (\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1(k\tau) J_0(l(\tau - \tau')) \\
& - \left[\left(\frac{v_\perp \cdot k_\perp}{k} \frac{v_\perp \cdot l_\perp}{l} + (v^\perp)^2 \frac{k_\perp \cdot l_\perp}{k l} \right) \delta^{ij} \delta^{mn} + \frac{k_\perp \cdot l_\perp}{k l} \varepsilon^{ij} \varepsilon^{mn} \right] J_1(k\tau) J_1(l(\tau - \tau'))
\end{aligned}$$

$$\begin{aligned}
v^\alpha v^\beta X^{\alpha\beta} = & \dots (v^\perp)^2 J_0(k\tau) J_0(l(\tau - \tau')) \delta^{ij} \delta^{mn} \\
& - \left(v_\perp^2 (\hat{k} \cdot \hat{l}) - (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \right) J_1(k\tau) J_1(l(\tau - \tau')) \varepsilon^{ij} \varepsilon^{mn} \dots \\
= & \dots (v^\perp)^2 J_0(k\tau) J_0(l(\tau - \tau')) \delta^{ij} \delta^{mn} \\
& - (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1(k\tau) J_1(l(\tau - \tau')) \varepsilon^{ij} \varepsilon^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} = & \dots (\delta^{ij} \delta^{mn} + v_\perp^2 \varepsilon^{ij} \varepsilon^{mn}) J_0(k\tau) J_0(l(\tau - \tau')) \\
& + i(v_\perp \cdot \hat{l})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0(k\tau) J_1(l(\tau - \tau')) \\
& + i(v_\perp \cdot \hat{k})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1(k\tau) J_0(l(\tau - \tau')) \\
& - \left[\left((v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) + (v^\perp)^2 (\hat{k} \cdot \hat{l}) \right) \delta^{ij} \delta^{mn} + (\hat{k} \cdot \hat{l}) \varepsilon^{ij} \varepsilon^{mn} \right] J_1(k\tau) J_1(l(\tau - \tau'))
\end{aligned}$$

$$\begin{aligned}
v^\alpha v^\beta X^{\alpha\beta} &= \dots (v^1)^2 J_0(K) J_0(L) \delta^{ij} \delta^{mn} \\
&\quad - \left(v_\perp^2 (\hat{k} \cdot \hat{l}) - (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \right) J_1(K) J_1(L) \varepsilon^{ij} \varepsilon^{mn} \dots \\
&= \dots (v^1)^2 J_0(K) J_0(L) \delta^{ij} \delta^{mn} \\
&\quad - (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1(K) J_1(L) \varepsilon^{ij} \varepsilon^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} &= \dots (\delta^{ij} \delta^{mn} + v_\perp^2 \varepsilon^{ij} \varepsilon^{mn}) J_0(K) J_0(L) \\
&\quad + i(v_\perp \cdot \hat{l})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0(K) J_1(L) \\
&\quad + i(v_\perp \cdot \hat{k})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1(K) J_0(L) \\
&\quad - \left[\left((v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) + (v^1)^2 (\hat{k} \cdot \hat{l}) \right) \delta^{ij} \delta^{mn} + (\hat{k} \cdot \hat{l}) \varepsilon^{ij} \varepsilon^{mn} \right] J_1(K) J_1(L)
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} &= \left[J_0(K) J_0(L) + i(v_\perp \cdot \hat{l}) J_0(K) J_1(L) + i(v_\perp \cdot \hat{k}) J_1(K) J_0(L) \right. \\
&\quad \left. - \left((v^1)^2 (\hat{k} \cdot \hat{l}) + (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \right) J_1(K) J_1(L) \right] \delta^{ij} \delta^{mn} \\
&\quad + \left[v_\perp^2 J_0(K) J_0(L) - i(v_\perp \cdot \hat{l}) J_0(K) J_1(L) - i(v_\perp \cdot \hat{k}) J_1(K) J_0(L) \right. \\
&\quad \left. - (\hat{k} \cdot \hat{l}) J_1(K) J_1(L) \right] \varepsilon^{ij} \varepsilon^{mn}
\end{aligned}$$

$$\begin{aligned}
v^\alpha v^\beta X^{\alpha\beta} &= \dots (v^1)^2 J_0^K J_0^L \delta^{ij} \delta^{mn} \\
&\quad - \left(v_\perp^2 (\hat{k} \cdot \hat{l}) - (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \right) J_1^K J_1^L \varepsilon^{ij} \varepsilon^{mn} \dots \\
&= \dots (v^1)^2 J_0^K J_0^L \delta^{ij} \delta^{mn} \\
&\quad - (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1^K J_1^L \varepsilon^{ij} \varepsilon^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} &= \dots (\delta^{ij} \delta^{mn} + v_\perp^2 \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_0^L \\
&\quad + i(v_\perp \cdot \hat{l})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_1^L \\
&\quad + i(v_\perp \cdot \hat{k})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1^K J_0^L \\
&\quad - \left[\left((v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) + (v^1)^2 (\hat{k} \cdot \hat{l}) \right) \delta^{ij} \delta^{mn} + (\hat{k} \cdot \hat{l}) \varepsilon^{ij} \varepsilon^{mn} \right] J_1^K J_1^L
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} = & \left[J_0^K J_0^L + i(v_\perp \cdot \hat{l}) J_0^K J_1^L + i(v_\perp \cdot \hat{k}) J_1^K J_0^L \right. \\
& - \left((v^1)^2 (\hat{k} \cdot \hat{l}) + (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \right) J_1^K J_1^L \Big] \delta^{ij} \delta^{mn} \\
& + \left[v_\perp^2 J_0^K J_0^L - i(v_\perp \cdot \hat{l}) J_0^K J_1^L - i(v_\perp \cdot \hat{k}) J_1^K J_0^L \right. \\
& \left. - (\hat{k} \cdot \hat{l}) J_1^K J_1^L \right] \varepsilon^{ij} \varepsilon^{mn}
\end{aligned}$$

$$\begin{aligned}
v^\alpha v^\beta X^{\alpha\beta} = & \dots (v^1)^2 J_0^K J_0^L \delta^{ij} \delta^{mn} \\
& - \left(v_\perp^2 (\hat{J}_1^K \cdot \hat{J}_1^L) - (v_\perp \cdot \hat{J}_1^K)(v_\perp \cdot \hat{J}_1^L) \right) \varepsilon^{ij} \varepsilon^{mn} \dots \\
= & \dots (v^1)^2 J_0^K J_0^L \delta^{ij} \delta^{mn} \\
& - (v_\perp \times \hat{J}_1^K)(v_\perp \times \hat{J}_1^L) \varepsilon^{ij} \varepsilon^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} = & \dots (\delta^{ij} \delta^{mn} + v_\perp^2 \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_0^L \\
& + i(v_\perp \cdot \hat{J}_1^L)(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0^K \\
& + i(v_\perp \cdot \hat{J}_1^K)(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0^L \\
& - \left[\left((v_\perp \cdot \hat{J}_1^K)(v_\perp \cdot \hat{J}_1^L) + (v^1)^2 (\hat{J}_1^K \cdot \hat{J}_1^L) \right) \delta^{ij} \delta^{mn} + (\hat{J}_1^K \cdot \hat{J}_1^L) \varepsilon^{ij} \varepsilon^{mn} \right]
\end{aligned}$$

$$\begin{aligned}
\delta^{\alpha\beta} X^{\alpha\beta} = & \left[J_0^K J_0^L + i(v_\perp \cdot \hat{J}_1^L) J_0^K + i(v_\perp \cdot \hat{J}_1^K) J_0^L \right. \\
& \left. - \left((v^1)^2 (\hat{J}_1^K \cdot \hat{J}_1^L) + (v_\perp \cdot \hat{J}_1^K)(v_\perp \cdot \hat{J}_1^L) \right) \right] \delta^{ij} \delta^{mn} \\
& + \left[v_\perp^2 J_0^K J_0^L - i(v_\perp \cdot \hat{J}_1^L) J_0^K - i(v_\perp \cdot \hat{J}_1^K) J_0^L \right. \\
& \left. - (\hat{J}_1^K \cdot \hat{J}_1^L) \right] \varepsilon^{ij} \varepsilon^{mn}
\end{aligned}$$

FULL TENSOR SIMPLIFIED (without integral/colour space envelope)

$$\begin{aligned}
X^{11} = & \dots \left[J_0^K J_0^L \right. \\
& + i(v_\perp \cdot \hat{l}) J_0^K J_1^L \\
& + i(v_\perp \cdot \hat{k}) J_1^K J_0^L \\
& \left. - (v_\perp \cdot \hat{l})(v_\perp \cdot \hat{k}) J_1^K J_1^L \right] \delta^{ij} \delta^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
X^{1\beta} = & \dots \left[-i v^1 \hat{l}^\beta J_0^K J_1^L \right. \\
& \left. + (v_\perp \cdot \hat{k}) v^1 \hat{l}^\beta J_1^K J_1^L \right] \delta^{ij} \delta^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
X^{\alpha 1} = & \dots \left[-i v^1 \hat{k}^\alpha J_1^K J_0^L \right. \\
& \left. + (v_\perp \cdot \hat{l}) v^1 \hat{k}^\alpha J_1^K J_1^L \right] \delta^{ij} \delta^{mn} \dots
\end{aligned}$$

$$\begin{aligned}
X^{\alpha\beta} = & \dots \left\{ \left[- \left(\delta^{\alpha\beta} - \frac{\hat{l}^\alpha \hat{k}^\beta}{\hat{l} \cdot \hat{k}} \right) (\hat{l} \cdot \hat{k}) J_1^K J_1^L \right. \right. \\
& - i \left(\delta^{\alpha\beta} - \frac{v^\alpha \hat{k}^\beta}{v_\perp \cdot \hat{k}} \right) (v_\perp \cdot \hat{k}) J_1^K J_0^L \\
& - i \left(\delta^{\alpha\beta} - \frac{\hat{l}^\alpha v^\beta}{\hat{l} \cdot v_\perp} \right) (\hat{l} \cdot v_\perp) J_0^K J_1^L \\
& + \left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v_\perp^2} \right) v_\perp^2 J_0^K J_0^L \left. \right] \varepsilon^{ij} \varepsilon^{mn} \\
& \left. - (v^1)^2 \hat{k}^\alpha \hat{l}^\beta J_1^K J_1^L \delta^{ij} \delta^{mn} \right\} \dots
\end{aligned}$$

$$\begin{aligned}
X^{\alpha\beta} = & \dots \left\{ \left[-\varepsilon^{\alpha\sigma} \hat{k}^\sigma \varepsilon^{\beta\rho} \hat{l}^\rho J_1^K J_1^L \right. \right. \\
& - i\varepsilon^{\alpha\sigma} \hat{k}^\sigma \varepsilon^{\beta\rho} v^\rho J_1^K J_0^L \\
& - i\varepsilon^{\alpha\rho} v^\rho \varepsilon^{\beta\sigma} \hat{l}^\sigma J_0^K J_1^L \\
& \left. + \varepsilon^{\alpha\sigma} v^\sigma \varepsilon^{\beta\rho} v^\rho J_0^K J_0^L \right] \varepsilon^{ij} \varepsilon^{mn} \\
& \left. - (v^1)^2 \hat{k}^\alpha \hat{l}^\beta J_1^K J_1^L \delta^{ij} \delta^{mn} \right\} \dots
\end{aligned}$$

$$\frac{dE}{dx} \text{ and } \hat{q}$$

$$\begin{aligned} \frac{dE}{dx} &= -\frac{v}{T} \frac{v^\alpha v^\beta}{v^2} X^{\alpha\beta} \\ &= \dots - \frac{v}{T} \frac{1}{v^2} \left[(v^1)^2 J_0^K J_0^L \delta^{ij} \delta^{mn} - (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1^K J_1^L \varepsilon^{ij} \varepsilon^{mn} \right] \dots \end{aligned}$$

Pure transverse movement: $v^1 = 0, v_\perp^2 = v^2$

$$\left(\frac{dE}{dx} \right)_\perp = \dots \frac{1}{Tv} (v_\perp \times \hat{k})(v_\perp \times \hat{l}) J_1^K J_1^L \varepsilon^{ij} \varepsilon^{mn} \dots$$

Pure longitudinal movement: $v^1 = v, v_\perp = 0$

$$\left(\frac{dE}{dx} \right)_\parallel = \dots - \frac{v}{T} J_0^K J_0^L \delta^{ij} \delta^{mn} \dots$$

$$\begin{aligned} \hat{q} &= \frac{2}{v} \left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2} \right) X^{\alpha\beta} \\ &= \dots \frac{2}{v} \left[\frac{v_\perp^2}{v^2} (\delta^{ij} \delta^{mn} + v^2 \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_0^L \right. \\ &\quad + i(v_\perp \cdot \hat{l})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_1^L \\ &\quad + i(v_\perp \cdot \hat{k})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1^K J_0^L \\ &\quad \left. - \left((v^1)^2 (\hat{k} \cdot \hat{l}) + (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \right) \left(\delta^{ij} \delta^{mn} + \frac{1}{v^2} \varepsilon^{ij} \varepsilon^{mn} \right) J_1^K J_1^L \right] \dots \end{aligned}$$

Pure transverse movement: $v^1 = 0, v_\perp^2 = v^2$

$$\begin{aligned} \hat{q}_\perp &= \dots \frac{2}{v} \left[(\delta^{ij} \delta^{mn} + v^2 \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_0^L \right. \\ &\quad + i(v_\perp \cdot \hat{l})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_0^K J_1^L \\ &\quad + i(v_\perp \cdot \hat{k})(\delta^{ij} \delta^{mn} - \varepsilon^{ij} \varepsilon^{mn}) J_1^K J_0^L \\ &\quad \left. - (v_\perp \cdot \hat{k})(v_\perp \cdot \hat{l}) \left(\delta^{ij} \delta^{mn} + \frac{1}{v^2} \varepsilon^{ij} \varepsilon^{mn} \right) J_1^K J_1^L \right] \dots \end{aligned}$$

Pure longitudinal movement: $v^1 = v, v_\perp = 0$

$$\hat{q}_\parallel = \dots - \frac{2}{v}(\hat{k} \cdot \hat{l}) \left(v^2 \delta^{ij} \delta^{mn} + \varepsilon^{ij} \varepsilon^{mn} \right) J_1^K J_1^L \dots$$